



Università
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Introduction

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

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What is Artificial Intelligence?

There is no universally accepted definition of Artificial Intelligence

A.I. is the endeavour of building an intelligent artifact

But... what is intelligence?

Some (early) definitions:

- ▶ It is the ability to learn (Buckingham, 1921)
- ▶ This faculty is judgment, otherwise called good sense, practical sense, initiative, the faculty of adapting one's self to circumstances (Binet and Simon, 1961)
- ▶ It is the ability to perform well in an intelligence test (Boring, 1961)

What is Artificial Intelligence?

More recently:

- ▶ A branch of computer science dealing with the simulation of intelligent behavior in computers. (Merriam-Webster)
- ▶ A program that can sense, reason, act, and adapt (Intel)
- ▶ Colloquially, the term 'artificial intelligence' is applied when a machine mimics 'cognitive' functions that humans associate with other human minds, such as 'learning' and 'problem solving'. (Wikipedia)

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Note the focus on **mimicking** human behavior.



Turing Test

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd.

(continues...)

Turing Test

Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

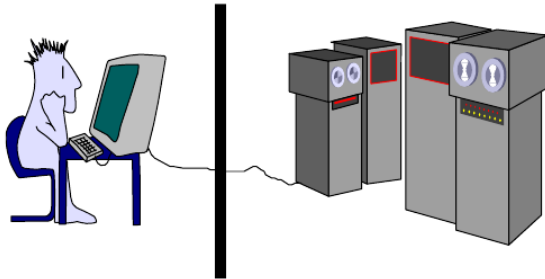
The new form of the problem can be described in terms of a game which we call the 'imitation game.' It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart front the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either "X is A and Y is B" or "X is B and Y is A." (*continues...*)

Turing Test

The interrogator is allowed to put questions to A and B thus: C: Will X please tell me the length of his or her hair? Now suppose X is actually A, then A must answer. It is A's object in the game to try and cause C to make the wrong identification. His answer might therefore be: "My hair is shingled, and the longest strands are about nine inches long."

In order that tones of voice may not help the interrogator the answers should be written, or better still, typewritten. The ideal arrangement is to have a teleprinter communicating between the two rooms. Alternatively the question and answers can be repeated by an intermediary. The object of the game for the third player (B) is to help the interrogator. The best strategy for her is probably to give truthful answers. She can add such things as "I am the woman, don't listen to him!" to her answers, but it will avail nothing as the man can make similar remarks. (*continues...*)

Turing Test



We now ask the question, "What will happen when a machine takes the part of A in this game?" Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, "Can machines think?"

Turing Test

To pass the test a machine must possess the following skills:

- ▶ Natural language elaboration (to interact with the interrogator)
- ▶ Knowledge representation (to memorize information before and during the dialogue)
- ▶ Automatic reasoning (to use the acquired knowledge to answer the question and draw conclusions)
- ▶ Learning (to adapt to new situations)

“Total” Turing Test

The machine can access an audio/video feed so that the interrogator can test its perception skills; further, the interrogator can pass objects to be manipulated. This requires:

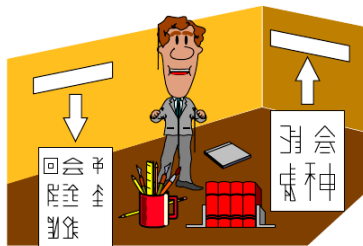
- ▶ Perception (to analyze and comprehend images and sounds)
- ▶ Robotics (to manipulate objects)

Weak AI: Can a machine exhibit intelligent behavior?

Strong AI: Can a machine have self awareness?

Searle's Chinese Room

The Chinese room is a thought experiment by John Searle (1980). In the experiment, Searle imagines himself in a room acting as a computer by manually executing a **program** that convincingly simulates the behavior of a native Chinese speaker. People outside the room slide Chinese characters under the door and Searle, to whom "Chinese writing is just so many meaningless squiggles", is able to **create sensible replies**, in Chinese, by following the instructions of the program; that is, by moving papers around.



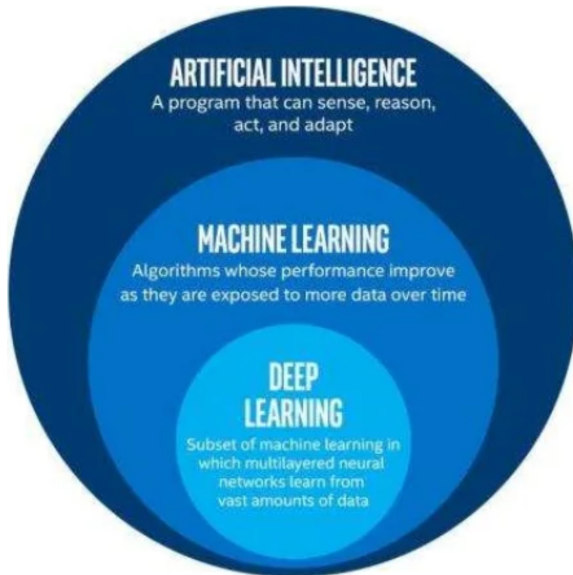
Searle's Chinese Room

Can Searle be said to understand Chinese in the same way that, as Searle says, "according to strong AI, [...] the appropriately programmed computer really is a mind, in the sense that computers given the right programs can be literally said to understand and have other cognitive states."

The experiment is the centerpiece of Searle's Chinese Room Argument which holds that a program cannot give a computer a "mind" or "understanding", regardless of how intelligently it may make it behave.

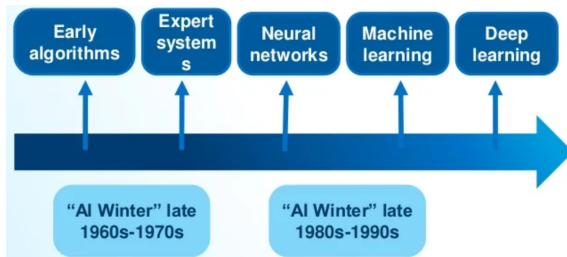


AI – Machine Learning – Deep Learning



History of AI

AI has experienced several hype cycles, where it has oscillated between periods of excitement and disappointment.



50s-60s: Early AI

- 1943: McCulloch and Pitts propose a model for an artificial neuron and analyze its properties
- 1949: Donald Hebb propose a learning mechanism in the brain, still of great interest
- 1950: Alan Turing developed the Turing Test
- 1950-53: Shannon and Turing work (independently) on chess playing programs
- 1951: Minsky and Edmonds develop the first “neural” computer
- 1956: Newell e Simon develop Logic Theorist
- 1956: Artificial Intelligence was accepted as a field at the Dartmouth Conference



Dartmouth Summer Project

A Proposal for the
DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug. 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

Proponents:

- J. McCarthy (Dartmouth College)
- M. L. Minsky (Harvard University)
- N. Rochester (IBM)
- C. E. Shannon (Bell labs.)



50s-60s: Early AI

1957: Frank Rosenblatt invented the perceptron algorithm

1957: First attempts at automatic translation

1958: LISP

1959: Arthur Samuel published an algorithm for a checkers program using machine learning

1961: Newell and Simon develop General Problem Solver (GPS)

First “AI Winter”

- 1966: ALPAC committee evaluated AI for automatic translation and determined there was little ROI
- 1969: Marvin Minsk published a book on the limitations of the perceptron
- 1972: Cook and Karp develop the computational complexity theory, showing that a lot of problems are intractable.
- 1973: The Lighthill report highlights AI's failure to live up to promises.

The two reports (ALPAC and Lighthill) resulted in the US government cutting funding for AI research.



80's AI Boom

Expert systems systems with programmed rules designed to mimic human experts. They were the first “widely-used” AI technology, with 2/3 of “Fortune 500” companies using them at their peak.

1986: Rumelhart, Hinton, and Williams introduce the “Backpropagation” algorithm which with multi-layer perceptrons can be trained.



Second Winter (late 80's – early 90's)

- ▶ Expert systems' progress slowed.
- ▶ Expert systems melded into software suites (SAP, Oracle, ...) to run on PCs
- ▶ Scaling problems with Neural Networks

Classical Machine Learning (late 90's – early 2000's)

- ▶ Developments in the SVM algorithm (Vapnik and Chervonenkis, 1963), particularly the integration of the “kernel trick” led to it becoming the machine learning method of choice.
- ▶ Machine Learning approaches had successes in speech recognition, medical diagnosis, robotics, and many other areas.
- ▶ the Deep Blue chess system beat Garry Kasparov
- ▶ Google search engine launched (using AI technology)

Rise of Deep Learning

- Neural Networks rebranded as **deep learning**

2006: Geoffrey Hinton published a paper on unsupervised pre-training that allowed deeper neural networks to be trained.

2012: AlexNet (Krizhevsky, Sutskever, and Hinton) achieved a top-5 error of 15.3% in the ImageNet 2012 Challenge, more than 10.8 percentage points lower than that of the runner up. This was made feasible due to the use of graphics processing units (GPUs) during training, an essential ingredient of the deep learning revolution